

Hackathon Day 2

Marketplace Technical Foundation

General E-commerce Marketplace (Furniture)

System Architecture Document Overview:

The system architecture for the furniture E-commerce marketplace integrates a dynamic frontend with a strong backend managed by Sanity CMS. Third-party APIs handle shipment tracking and payment processing. This modular design ensures the marketplace is scalable and provides a seamless user experience.

Components:

1: Frontend (Next.js):

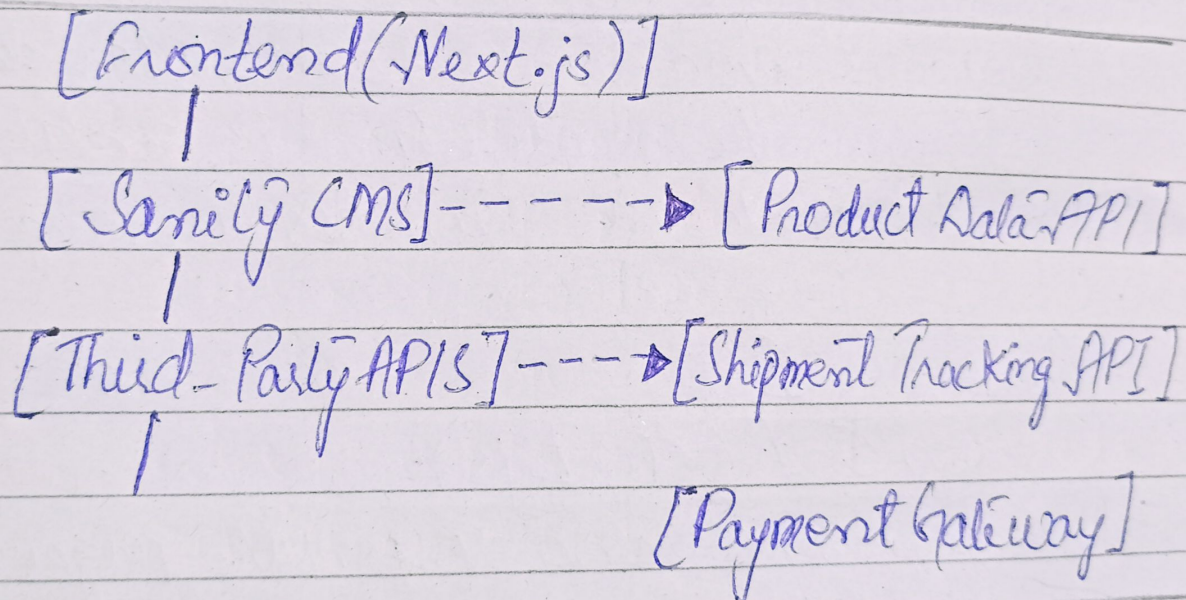
- Dynamic and responsive pages for browsing furniture, managing the cart, and completing the checkout process.
- Communication with the backend APIs to provide real-time data updates.

2: Sanity CMS:-

- Manages product, customer, and order information.

- Supports custom schemas to organize data flexibly.
- 3: Third-Party APIs:
- Shipment Tracking API: Provides real-time updates on the delivery status of orders.
 - Payment Gateway API: Ensures secure processing of payments.

Architecture Diagram:



Data Flow:

- 1: Users interact with the frontend to browse furniture and place orders.
- 2: The frontend makes API requests to Sanity CMS to fetch relevant data.
- 3: Orders are recorded in Sanity CMS, and updates are retrieved from third-party

APIs.

Date

4: Real-time updates on shipment status and payment processing are displayed to users.

API Specification Document
API Endpoints:

Endpoint	Method	Purpose	Example Payload/Response
/products	GET	Fetch all furniture product details	Response: { "id": 1, "name": "Sofa", "price": 200 }
/orders	POST	Create a new furniture order	Payload: { "customer-id": 123, "products": [...] }
/shipment	GET	Track order shipment status	Response: { "shipmentId": 456, "status": "Delivered" }
/loyalty-points	GET	Fetch loyalty points for a customer	Response: { "customer-id": 123, "points": 500 }

Example Endpoint Details:

1: /products

- Method: GET

- Description: Fetches all available furniture products.

- Response: `{ "id": 1, "name": "Sofa", "price": 200, "stock": 30 }`

2: /orders:

- Method: POST
- Description: Creates a new furniture order in the system.
- Payload: `{ "customerId": 123, "products": [ { "productId": 1, "quantity": 2 } ] }`
- Response: `{ "orderId": 789, "status": "Pending" }`

Workflow Diagram

Workflows:

1: User Registration:

- User signs up → Data stored in Sanily CMS → Confirmation email sent.

2: Product Browsing:

- User searches for furniture → Sanily CMS provides product data → Results displayed dynamically.

3: Order Placement:

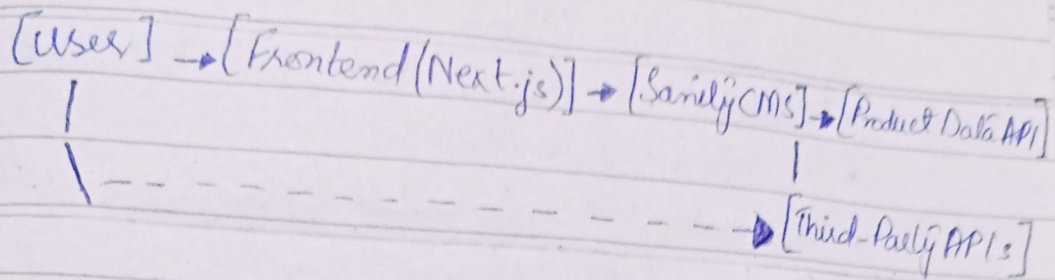
- User add items to cart → Proceeds to checkout → Order details saved in Sanily CMS → Payment processed.

4: Shipment Tracking:

- Shipment API fetches status → Updates displayed on user dashboard.

Diagram:

Date



A visual representation of user interactions and data flow within the system will illustrate the steps above.

Data Schema Design:

Entities and Attributes:

1: Products:

- Product ID
- Name
- Description
- Price
- Stock level
- Category (e.g., living Room, Bedroom, Office)
- Sub-Category (e.g., Sofa, Table, Chair)
- Images
- Tags (e.g., "New Arrival", "Discount")

2: Orders:

- Order ID
- Customer ID

- Product List (Product ID, Quantity)
- Total Amount
- Status (Pending, Shipped, Delivered)
- Payment Method
- Timestamps (Order Date, Expected Delivery Date)

3: Customers:

- Customer ID
- Name
- Email
- Address
- Contact Number
- Order History
- Preference (e.g., Style, Material, Color, Preferences)

4: Shipment:

- Shipment ID
- Order ID
- Delivery Status
- Courier Partner
- Expected Delivery Date

5: Delivery Zones:

- Zone ID
- Area Name
- Coverage (Cities / Regions)

6: Loyalty Points:

- Points ID
- Customer ID
- Points Earned
- Points Redeemed
- Expiry Date

Technical Roadmap

Phase 1: Initial Setup

1. Define business goals and target audience.
2. Draft data schema tailored to the furniture marketplace.
3. Set up Sanity CMS for backend management.

Phase 2: System Architecture and API Development

1. Design system architecture and workflows.
2. Develop API endpoints for products, orders, and shipments.
3. Integrate third-party APIs for shipment tracking and payment processing.

Phase 3: Frontend Development

1. Build responsive, user-friendly

pages using Next.js.

2. Implement real-time data fetching from APIs.

3. Develop dynamic features like search filters and personalized recommendations for furniture.

Phase 4: Testing and Deployment

1. Conduct unit and integration testing for APIs and frontend components.

2. Deploy the application on a scalable hosting platform.

3. Monitor performance and resolve any issues.

Phase 5: Post-Launch Optimization

1. Gather user feedback and make iterative improvements.

2. Introduce new features such as seasonal collection, discounts, and advanced loyalty programs.

3. Scale infrastructure to handle increased traffic and transactions.